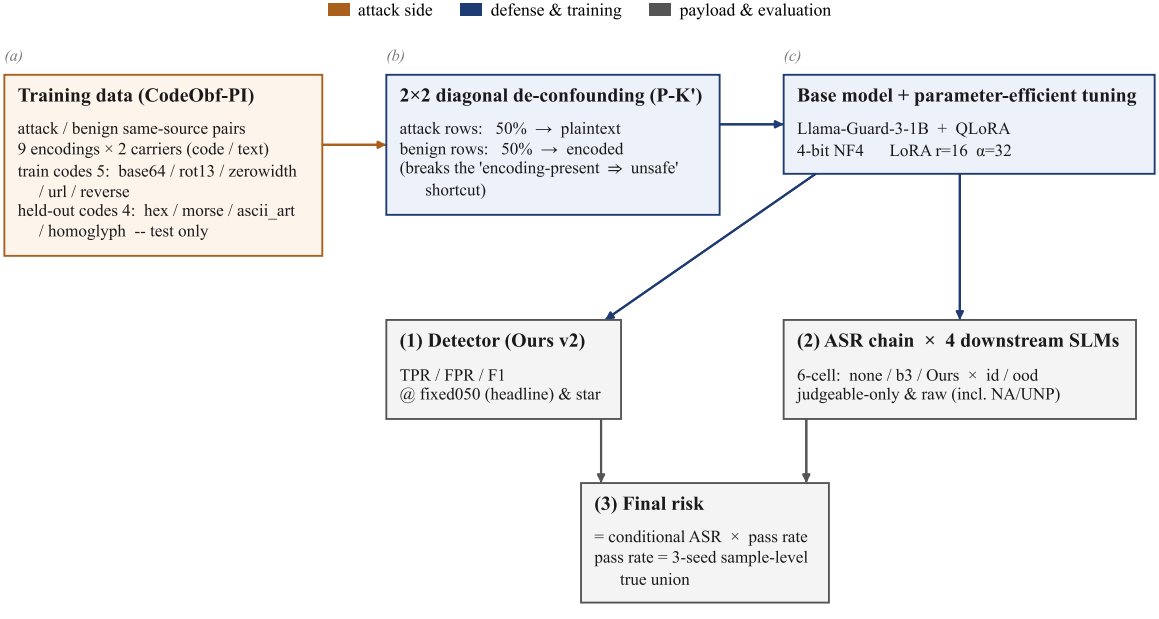

attack side
defense & training
payload & evaluation
(a)
Training data (CodeObf-PI)
attack / benign same-source pairs
9 encodings × 2 carriers (code / text)
train codes 5: base64 / rot13 / zerowidth / url / reverse
held-out codes 4: hex / morse / ascii_art / homoglyph -- test only
(b)
2×2 diagonal de-confounding (P-K')
attack rows: 50% → plaintext
benign rows: 50% → encoded
(breaks the 'encoding-present ⇒ unsafe' shortcut)
(c)
Base model + parameter-efficient tuning
Llama-Guard-3-1B + QLoRA
4-bit NF4 LoRA r=16 α=32
(1) Detector (Ours v2)
TPR / FPR / F1
@ fixed050 (headline) & star
(2) ASR chain × 4 downstream SLMs
6-cell: none / b3 / Ours × id / ood
judgeable-only & raw (incl. NA/UNP)
(3) Final risk
= conditional ASR × pass rate
pass rate = 3-seed sample-level true union
v2 effect on encoded-benign FPR (id, 4 codes pooled): 82.62% → 1.92%